

```
!pip install transformers datasets faiss-cpu sentence-transformers torch
```

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Requirement already satisfied: pyyaml<=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
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Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0) (2025.3.0)
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Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
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Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3) (1.2.0)
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Requirement already satisfied: MarkupSafe<=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch) (3.0.3)
Requirement already satisfied: python-dateutil<=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.8.2)
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Requirement already satisfied: joblib<=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->sentence-transformers) (1.2.0)
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Requirement already satisfied: h11<=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0) (0.16.0)
Requirement already satisfied: six<=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil<=2.8.2) (1.16.0)
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Downloading faiss_cpu-1.13.2-cp310-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (23.8 MB)
23.8/23.8 MB 81.6 MB/s eta 0:00:00

Installing collected packages: faiss-cpu
Successfully installed faiss-cpu-1.13.2
```

```
from transformers import AutoTokenizer, AutoModelForQuestionAnswering, pipeline

# Load DistilBERT fine-tuned for QA
model_name = "distilbert-base-uncased-distilled-squad"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForQuestionAnswering.from_pretrained(model_name)

qa_pipeline = pipeline("question-answering", model=model, tokenizer=tokenizer)

# Example context
context = """
Garima Kumari is a second-year Economics student at Miranda House, Delhi University.
She is passionate about software and product development, especially AI-driven projects.
"""

question = "Where does Garima study?"

result = qa_pipeline(question=question, context=context)
print("Answer:", result['answer'])
```

```

config.json: 100%                                451/451 [00:00<00:00, 15.3kB/s]
tokenizer_config.json: 100%                        48.0/48.0 [00:00<00:00, 1.55kB/s]
vocab.txt: 100%                                   232k/232k [00:00<00:00, 2.64MB/s]
tokenizer.json: 100%                              466k/466k [00:00<00:00, 1.04MB/s]
model.safetensors: 100%                          265M/265M [00:04<00:00, 135MB/s]
Loading weights: 100%                            102/102 [00:00<00:00, 308.19it/s, Materializing param=qa_outputs.weight]
Answer: Miranda House, Delhi University

```

```

from sentence_transformers import SentenceTransformer, util
import torch

# Sample FAQ
faq = {
    "Where are you located?": "We are located in New Delhi, India.",
    "What courses do you offer?": "We offer courses on AI, Economics, and Software Development.",
    "How can I contact support?": "You can email support@example.com."
}

# Embed questions
model = SentenceTransformer('all-MiniLM-L6-v2')
faq_questions = list(faq.keys())
faq_embeddings = model.encode(faq_questions, convert_to_tensor=True)

def faq_bot(query):
    query_embedding = model.encode(query, convert_to_tensor=True)
    similarity = util.cos_sim(query_embedding, faq_embeddings)
    best_idx = torch.argmax(similarity)
    return faq[faq_questions[best_idx]]

# Example
print(faq_bot("How do I reach support?"))

```

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modules.json: 100%                                349/349 [00:00<00:00, 16.4kB/s]
config_sentence_transformers.json: 100%            116/116 [00:00<00:00, 3.98kB/s]
README.md: 10.5k/? [00:00<00:00, 884kB/s]
sentence_bert_config.json: 100%                   53.0/53.0 [00:00<00:00, 1.33kB/s]
config.json: 100%                                612/612 [00:00<00:00, 17.8kB/s]
model.safetensors: 100%                          90.9M/90.9M [00:00<00:00, 118MB/s]
Loading weights: 100%                            103/103 [00:00<00:00, 199.51it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+--
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100%                        350/350 [00:00<00:00, 12.8kB/s]
vocab.txt: 232k/? [00:00<00:00, 2.58MB/s]
tokenizer.json: 466k/? [00:00<00:00, 7.21MB/s]
special_tokens_map.json: 100%                     112/112 [00:00<00:00, 13.3kB/s]
config.json: 100%                                190/190 [00:00<00:00, 16.5kB/s]
You can email support@example.com.

```

